

Juan David Muñoz-Bolaños

Rechenauerstraße 42, Innsbruck, Austria (Tirol)
+43 676 9115145
juan.munoz@i-med.ac.at

Leica Microsystems CMS GmbH

Hiring Team
Am Friedensplatz 3
68165 Mannheim, Germany

Subject: Application for System Developer Confocal Microscopy (d/f/m)

Dear Hiring Team,

I am writing to express my strong enthusiasm for the System Developer Confocal Microscopy position. Concluding my PhD in Innsbruck, my hands-on expertise designing and characterizing advanced nonlinear microscopy platforms directly aligns with your need for innovation in core confocal technologies.

In my current role, I take sole responsibility for the architecture of complex Two-Photon and Confocal microscopy setups. I routinely integrate critical components: high-power pulsed lasers, spatial light modulators, and sensitive PMTs/spectrometers to correct severe aberrations, heavily mirroring the hardware challenges of point-scanning microscope development.

Advanced Microscopy & Prototyping:

During my PhD and my work at IPHT Jena, I have specialized in building optical prototypes from the ground up. I am highly comfortable evaluating new core concepts, optimizing laser coupling, and pushing signal-to-noise ratios in photon-starved imaging regimes.

System Architecture & Control:

Modern microscopy requires a seamless bridge between physics and electronics. I develop high-performance Python pipelines and hardware-control software to synchronize laser excitation with real-time detection, ensuring robust system performance analogous to industrial R&D.

Project Leadership & Problem Solving:

From academic research to industrial machine vision at INTECOL, I utilize a highly structured, analytical approach to problem-solving. I am used to managing complex project deliverables independently while enjoying in interdisciplinary teams.

I hold a strong working proficiency in German (B2) alongside fluent English. I welcome the opportunity to discuss how my hands-on experience in multiphoton optical prototyping can directly contribute to Leica's advanced microscopy development.

Sincerely,

Juan David Muñoz-Bolaños

Optical System Engineer
Medical University of Innsbruck

Juan David Muñoz-Bolaños

Optical System Engineer | Adaptive Optics

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juan.munoz@i-med.ac.at | [LinkedIn Profile](#)



PROFESSIONAL SUMMARY

Optical Engineer specialized in the architecture, integration, and prototyping of advanced nonlinear microscopy platforms. Extensive hands-on experience in Two-Photon and Confocal microscopy, high-power laser integration, and precise metrology evaluation. Proven capability to bridge optical physics with hardware control software to drive R&D from conceptualization to reliable laboratory prototypes.

TECHNICAL COMPETENCIES

Advanced Microscopy & Optical Architecture

Substantial expertise in Two-Photon, confocal, and custom nonlinear microscopy principles. Demonstrated ability to design, assemble, and optimize complex optical platforms that push the physical limits of diffraction and deep tissue scaling.

Prototyping & Core Technologies

Hands-on integration of high-power continuous-wave/pulsed lasers, spatial light modulators, acousto-optics, and highly sensitive detection modules (PMTs, Spectrometers). Focus on signal-to-noise optimization.

Software Integration & Control

Advanced Python (JAX, NumPy) for hardware interfacing, data acquisition, and signal processing. Experience synchronizing scanning technologies with real-time detection software pipelines.

PROFESSIONAL EXPERIENCE

Medical University of Innsbruck

Innsbruck, Austria

PhD Researcher - Optical Engineering

Jan 2022 – Present

- **Microscopy System Architecture:** Designed and constructed advanced Two-Photon microscopy setups from the ground up, directing the full optical integration from laser source to detector.
- **Core Technology Integration:** Integrated and characterized novel core technologies, including adaptive optics hardware, to fundamentally improve spatial resolution in deep, scattering media.
- **Project Management:** Led independent R&D initiatives, taking sole operational responsibility for evaluating new concepts and maintaining complex laboratory prototypes.

Leibniz Institute of Photonic Technology (IPHT)

Jena, Germany

R&D Assistant Researcher

Jun 2020 – Dec 2021

- **Prototype Construction:** Assembled and calibrated vibrational point-scanning imaging prototypes, integrating fiber probes alongside sensitive detection setups.
- **Innovation & Evaluation:** Bridged theoretical R&D with practical lab implementation, utilizing structured data evaluation to streamline experimental characterization.

INTECOL S.A.S.

Medellin, Colombia

Machine Vision System Engineer

Mar 2018 – Mar 2019

- **System Reliability:** Ensured strict industrial robustness for optical inspection setups, gaining crucial experience in lifecycle management outside of academic settings.
- **Interdisciplinary Teamwork:** Collaborated heavily with external suppliers and cross-functional teams to integrate complex machine vision tracking systems.

EDUCATION

Medical University of Innsbruck
Ph.D. in Adaptive Optics & Microscopy

Austria
Jan 2022 – Present

Friedrich Schiller University Jena
M.Sc. in Photonics

Germany
2019 – 2021

University of Cauca
B.Sc. in Physics Engineering

Colombia
2012 – 2018

PUBLICATIONS & AWARDS

- **Awards:** ZEISS Summer School (2025), COLFUTURO Grant (2019).
- Muñoz-Bolaños, J. D., et al. Design of a dispersive 1064 nm fiber probe Raman imaging spectrometer and its application to human bladder resectates. *Applied Sciences*, 14(11), 4726 (2024).
- Muñoz-Bolaños, J. D. et al. Confocal Raman Microscopy with Adaptive Optics. *ACS Photonics* 12(1), 176–184 (2025).
- Sohmen, M., Muñoz-Bolaños, J. D., et al. Optofluidic adaptive optics in multi-photon microscopy. *Biomedical Optics Express* 14(4), 1562–1578 (2023).

LANGUAGES

English (Proficient),
German (Intermediate/Working Proficiency),
Spanish (Native)

REFERENCES

Prof. Monika Ritsch-Marte
Head of Institute for Biomedical Physics
monika.ritsch-marte@i-med.ac.at

Medical University of Innsbruck

Dr. Christoph Krafft
Group leader: Spectroscopy Imaging
christoph.krafft@leibniz-ipht.de

Leibniz Institute of Photonic Technology



URKUNDE

Die Friedrich-Schiller-Universität Jena verleiht durch die
Physikalisch-Astronomische Fakultät
mit dieser Urkunde

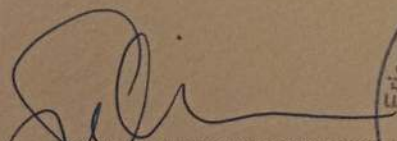
Herrn Juan David Munoz Bolanos
geb. 04.06.1994 in La Cruz (Kolumbien)
den Hochschulgrad

MASTER OF SCIENCE (M.Sc.)

Er hat die Masterprüfung (120 ECTS)
im Studiengang »Photonics«

bestanden.

Jena, 20.12.2021


Prof. Dr. Christian Spielmann
Dekan




Prof. Dr. Silvana Botti
Vorsitzende des
Prüfungsausschusses

Passaporto № / Passport No.

MUNOZ BOLANOS

COLOMBIANA

Sexo / Sex Lugar de nacimiento / Place of birth

Fecha de expedición / Date of issue

11 DIC/DEC 2018
Fecha de vencimiento / Date of expiry

10 DIC/DEC 2028

Autocidad / Authority

G. ANTIOQUIA

Firma del titolare / Legittimo rappresentante

Dear Family

P<COLMUNOZ<BOLANOS<<JUAN<DAVID<<<<<<<<<<<<<<<
AV410884<OCOL9406041M2812106CC1061772278<<56

Zertifikat

Muñoz-Bolaños JUAN DAVID

geboren am 04.06.1994

in La Cruz / Kolumbien

hat das Modul **Schriftliche Prüfung**

ÖSD Zertifikat B2

am Prüfungszentrum

BFI Tirol Bildungs GmbH

in Innsbruck / Österreich

bestanden



Datum: 16.09.2025

für das ÖSD: Prüfungsvorsitzende/r
Maria VALLAZZA



ÖSD ist Mitglied von



ö s d

österreich schweiz deutschland

ID-Nummer: ZB2Ö25s34291

Juan Munoz Bolanos

has participated in the

**ZEISS European
Summer School**

Lithography Optics &
Semiconductor Technologies

26 - 27 June 2025

ZEISS Semiconductor Manufacturing Technology, Oberkochen, Germany

Thomas Marschner

Dr. Thomas Marschner
Program Manager Funding Projects

Thomas Stämmler

Dr. Thomas Stämmler
CTO & Head of Product Strategy



INTECOL

OPTIMIZING THE RHYTHM OF THE WORLD

Medellin, February 4, 2019

TO WHOM IT MAY CONCERN

Integración Tecnológica Industrial Intecol S.A.S. with Tax ID (NIT) 811.041.410-4 certifies that Mr. JUAN DAVID MUÑOZ BOLAÑOS, identified with Citizenship ID No. 1.061.772.278, has been employed by this company since July 3, 2018, under an indefinite term contract, serving in the position of Junior Project Engineer;

earning a salary of One million five hundred thousand pesos (\$1,500,000) plus three hundred thousand pesos (\$300,000) mobility allowance.

I will gladly address any additional information at the phone number 316 3322.

We appreciate your attention,

INTECOL

NIT. 811.041.410-4 - Tel.: 3163822

LENY ADRIANA RUIZ PINO

Administrative and Financial Director

Integración Tecnológica Industrial Intecol S.A.S.
www.intecol.com.co
Medellin Calle 42 #63c-82
PBX. (574) 316 3322 Cell: (57) 301 430 2532

AGREEMENT FOR GUESTS to carry out a practical training with

Mr. Juan David Munoz Bolanos **Date of birth: 1994-06-04**
temporarily living at Emil-Wölk-Str. 7, 07747 Jena **(guest),**

Student of the Friedrich-Schiller-University Jena, Abbe School of Photonics, gets in execution of his practical course in the research department Spectroscopy and Imaging, working group Raman and Infrared Histopathology (0104) from 2021-01-01 up to 2021-09-30 admission to the necessary rooms within Leibniz-IPHT.

The use of further devices, media and other infrastructure takes place in agreement with the Heads of the respective departments.

Mr. Juan David Munoz Bolanos has to observe the security regulations for working and other regulations valid in Institute as well as instructions of the laboratory and clean room responsible persons of the Leibniz-IPHT.

Mr. Juan David Munoz Bolanos has to treat all affairs and procedures strictly confidential. Copies, duplicates and others of business and technical information of each kind are forbidden, unless these are sanctioned by the responsible Department Heads.

Publications, reports and lectures about topics which belong together with the fields of work of Leibniz-IPHT require the previous written agreement of the Executive Board.

The liability of Mr. Juan David Munoz Bolanos is limited to cases of wilful action and gross negligence.

Mr. Juan David Munoz Bolanos is insured against accidents on the Friedrich-Schiller-University Jena, Abbe School of Photonics in the context of his/her practical course during his/her length of stay at Leibniz-IPHT.

Jena, 13 November 2020


Prof. Dr. Jürgen Popp
Chairman
of Leibniz-IPHT


F. Sondermann
Executive Member
of Leibniz-IPHT


Juan David Munoz Bolanos
Guest

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